

JEE MAIN REVISION MASTER: MATHEMATICAL INDUCTION OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Mathematical Induction — Core Principles & Governing Laws	3 PYQs	Core Principle
Standard	Mathematical Induction — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis
Standard	Mathematical Induction — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mathematical Induction — Core Principles & Governing Laws	$=1+3+5...+(2K-1)=3+K^2, \quad a_n=\sqrt{7+a_n}, \quad a_n=\sqrt{7+\sqrt{7+\sqrt{7+.....}}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mathematical Induction — Core Principles & Governing Laws (Moderate Yield — 3 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Mathematical Induction.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Mathematical Induction.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Mathematical Induction PYQs.

Representative Source PYQs:

- [2005 Shift] Q#1 (MCQ): "If $A = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}$, then which one of the followin..." [Ans: N/A]
- [2004 Shift] Q#2 (MCQ): "Let $S(K) = 1+3+5...+(2K-1)=3+K^2$. Then which of the following is true..." [Ans: N/A]
- [2002 Shift] Q#3 (MCQ): "If $a_n = \sqrt{7+\sqrt{7+\sqrt{7+.....}}}$ having n radical signs then by methods of mathematical induction which is true..." [Ans: N/A]

3.2 Mathematical Induction — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Mathematical Induction.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2005 Shift] Q#1 (MCQ): "If $A = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}$, then which one of the followin..." [Ans: N/A]
- [2004 Shift] Q#2 (MCQ): "Let $S(K) = 1+3+5+\dots+(2K-1) = 3+K^2$. Then which of the following is true..." [Ans: N/A]

3.3 Mathematical Induction — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Mathematical Induction.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2005 Shift] Q#1 (MCQ): "If $A = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}$, then which one of the followin..." [Ans: N/A]
- [2004 Shift] Q#2 (MCQ): "Let $S(K) = 1+3+5+\dots+(2K-1) = 3+K^2$. Then which of the following is true..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: AIEEE 2005 (Q#1)

If $A = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction?

- (A) $A^n = nA - (n-1)I$
- (B) $A^n = 2^{n-1}A - (n-1)I$
- (C) $A^n = nA + (n-1)I$
- (D) $A^n = 2^{n-1}A + (n-1)I$

Correct Answer: None

Solution Analysis: Given $A = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix}$ $\therefore A \times A = A^2 = \begin{bmatrix} 1 & 0 & 2 & 1 \end{bmatrix}$ and $A^3 = A^2 \times A = \begin{bmatrix} 1 & 0 & 3 & 1 \end{bmatrix}$ So we can say $A^n = \begin{bmatrix} 1 & 0 & n & 1 \end{bmatrix}$ Now $nA - (n-1)I = \begin{bmatrix} n & 0 & n & n \end{bmatrix} - \begin{bmatrix} n-1 & 0 & 0 & n-1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & n & 1 \end{bmatrix} = A^n \therefore A^n = nA - (n-1)I$